Data Wrangling I

The questions below refer to the data frame called `day_at_cal`, with the habits of six students during a day at Cal including the day of the week (`weekday`), their study spot that day (`study_spot`), the number of total hours that they used AI in the course of their work (`ai`), and the number of steps walked, in thousands (`steps_k`). To check your answers in R, you can copy and paste the code that creates this data frame from the last slide from class.

```
day_at_cal
```

```
# A tibble: 6 × 4
  weekday study_spot ai_hours steps_k
  <ord>   <fct>         <dbl>   <dbl>
1 Mon     Moffitt         1.2     8.5
2 Mon     Doe             0.4     6.2
3 Tue     Cory            2.1     9.1
4 Tue     MLK             1       7
5 Wed     CITRIS          0.8     5.8
6 Wed     Soda            1.6    10.3
```

Write a single `dplyr` command that will produce each of the following.

1. A data frame containing the 2nd and 5th students.

2. A data frame without the 3rd student.

3. A data frame with only the AI hours column.

4. A data frame with only the factor columns[1].

5. A data frame containing only the students with at least one hour of AI use.

6. A data frame with students who study in either Moffitt or Doe.

7. A data frame with a fifth column called `ai_min` with the AI use in minutes.

8. A data frame with a fifth (logical) column called `active_day` where the number of steps exceeds 8,000.

[1]There is a clever way to do this that avoids having to type the names of the columns that are factors. See the help file for `select()`, particularly about predicate functions.

9. The same data frame sorted in descending order of number of hours of AI.

10. The same data frame sorted according to the day of the week. Within rows with the same day of the week, sort in descending order of the number of steps.

11. A data frame with the overall mean hours of AI use. Name that statistic `mean_ai`.

12. A data frame with the minimum steps walked (`min_steps`), the maximum steps walked (`max_steps`), and the standard deviation of steps walked (`sd_steps`).

13. Skim the documentation for the `tibble` package (https://tibble.tidyverse.org/) then write down three ways in which tibbles behave differently than data frames.

14. If you're interested in reading more about the design philosophy behind the Tidyverse, read *The Tidy Tools Manifesto* by Hadley Wickham, https://cran.r-project.org/web/packages/tidyverse/vignettes/manifesto.html, (it's just two pages).

Data Wrangling II

For this problem set, you'll be working with the `storms` tibble built into `dplyr`. This is a larger data frame than previous, so you'll need to inspect it more carefully in R in order formulate the necessary code below.

```r
library(dplyr)
storms
```

```
# A tibble: 19,537 × 13
   name   year month   day  hour   lat  long status      category  wind pressure
   <chr> <dbl> <dbl> <int> <dbl> <dbl> <dbl> <fct>          <dbl> <int>    <int>
 1 Amy    1975     6    27     0  27.5 -79   tropical d…       NA    25     1013
 2 Amy    1975     6    27     6  28.5 -79   tropical d…       NA    25     1013
 3 Amy    1975     6    27    12  29.5 -79   tropical d…       NA    25     1013
 4 Amy    1975     6    27    18  30.5 -79   tropical d…       NA    25     1013
 5 Amy    1975     6    28     0  31.5 -78.8 tropical d…       NA    25     1012
 6 Amy    1975     6    28     6  32.4 -78.7 tropical d…       NA    25     1012
 7 Amy    1975     6    28    12  33.3 -78   tropical d…       NA    25     1011
 8 Amy    1975     6    28    18  34   -77   tropical d…       NA    30     1006
 9 Amy    1975     6    29     0  34.4 -75.8 tropical s…       NA    35     1004
10 Amy    1975     6    29     6  34   -74.8 tropical s…       NA    40     1002
# ℹ 19,527 more rows
# ℹ 2 more variables: tropicalstorm_force_diameter <int>,
#   hurricane_force_diameter <int>
```

15. What are the dimensions of the data frame? What does every row correspond to?

16. Calculate the average wind speed and pressure for each category of storm.

17. Find the number of observations and the maximum wind speed for each combination of storm status and category. Sort the results by maximum wind speed in descending order[2].

[2]Hint 1: `n()` is a function that returns the row count when used inside `summarize()`. Hint 2: You can group by more than one variable.

18. Use the `.by` shortcut to calculate the average pressure for each year. Repeat the calculation using `group_by()`. Do you get the same answer?

19. For each storm name, calculate the z-score of wind speed (standardized within each storm). Add a new column called `wind_z` that constains these standarized values.

Additional advanced exercises coming soon.

Intro to Data Visualization

Coming soon!